

C.MAN

1) MACs per layer

Formula :- $\text{MACs} = \text{Input} \times \text{Output}$

$$\text{Layer 1} :- 784 \times 256 = \underline{200,704}$$

$$\text{Layer 2} :- 256 \times 128 = \underline{32,768}$$

$$\text{Layer 3} :- 128 \times 10 = \underline{1,280}$$

2) Total MACs.

$$200,704 + 32,768 + 1,280 = \underline{234,752}$$

3) Total Parameters.

$$\text{Same as weights} \\ = \underline{234,752}$$

4) weight Memory (FP 32)

$$234,752 \times 4 = \underline{939,008 \text{ bytes}}$$

5) Activation Memory

Store :- Input = 784, Hidden1 = 256, Hidden2 = 128, output = 10

$$784 + 256 + 128 + 10 = 1,178 \text{ bytes}$$

$$= 1,178 \times 4 = \underline{4,712 \text{ bytes}}$$

6) Arithmetic Intensity.

$$\text{Formula} = \frac{2 \times \text{MACs}}{\text{weight bytes} + \text{Activation bytes}}$$

$$= \frac{2 \times 234,752}{939,008 + 4,712} = \frac{469,504}{943,720} = \underline{20.497 \text{ FLOP/byte}}$$